

Ex. No: 09	ACCESSING APIs THROUGH HTTP
Date of Exercise	22.09.2026
Git-hub Host Link	

Aim:

To access a REST API using HTTP and retrieve and display JSON data in a Flutter application.

Description:

This experiment demonstrates how Flutter communicates with a REST API using the http package. The application sends a GET request to the JSONPlaceholder API, converts the JSON response into Dart data, and displays the posts using ListView.builder. Loading and error states are also handled.

Procedure:

Step 1 : Create Flutter Project - flutter create api_app

Step 2 : Add HTTP Package - Add http under dependencies in pubspec.yaml, then run:

Step 3 : Import Packages dart:convert for JSON decoding and the http package for API requests.

Step 4 : Create API Function Create a function using http.get() to retrieve data from the JSONPlaceholder REST API.

Step 5 : Decode JSON Data check the response status and convert the JSON response into Dart data using jsonDecode().

Step 6 : Display API Data Store the retrieved posts in a list and display the title and body of each post using ListView.builder().

Step 7 : Add Loading Display a loading indicator while the API data is being fetched.

Step 8 : Display an appropriate error message when the API request is unsuccessful

Step 9 : Test the Application Run the application in Chrome and verify that the API posts are displayed correctly.

Step 10 : Deploy to GitHub Pages - Build and upload the Flutter web application to the GitHub repository exp9-api-flutter and enable GitHub Pages using GitHub Actions.

Program:

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:http/http.dart' as http;

void main() {
  runApp(const ApiApp());
}

class ApiApp extends StatelessWidget {
  const ApiApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'API App',
      theme: ThemeData(
        scaffoldBackgroundColor: const Color(0xFFFF8F4FC),
        colorScheme: ColorScheme.fromSeed(
          seedColor: const Color(0xFF9C7AC7),
        ),
        appBarTheme: const AppBarTheme(
          backgroundColor: Color(0xFFBFA2DB),
          foregroundColor: Colors.white,
          centerTitle: true,
          elevation: 2,
        ),
        cardTheme: const CardThemeData(
          color: Color(0xFFF0E6F7),
          elevation: 2,
          margin: EdgeInsets.symmetric(
            horizontal: 12,
            vertical: 7,
          ),
        ),
      ),
      home: const ApiHomePage(),
    );
  }
}

class ApiHomePage extends StatefulWidget {
  const ApiHomePage({super.key});

  @override
```

```
State<ApiHomePage> createState() => _ApiHomePageState();
}
```

```
class _ApiHomePageState extends State<ApiHomePage> {
  List<dynamic> posts = [];
  bool isLoading = true;
  String errorMessage = "";
```

```
@override
void initState() {
  super.initState();
  fetchData();
}
```

```
Future<void> fetchData() async {
  setState(() {
    isLoading = true;
    errorMessage = "";
  });
```

```
  try {
    final response = await http.get(
      Uri.parse(
'https://jsonplaceholder.typicode.com/posts',
    ),
  );
```

```
  if (response.statusCode == 200) {
    final data = jsonDecode(response.body);
```

```
    setState(() {
      posts = data;
      isLoading = false;
    });
  } else {
    setState(() {
      errorMessage = 'Failed to load data';
      isLoading = false;
    });
  }
  } catch (e) {
    setState(() {
      errorMessage = 'Network error. Please try again.';
    });
  }
}
```

```
isLoading = false;
});
}
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text(
        'REST API Posts',
        style: TextStyle(
          fontWeight: FontWeight.bold,
        ),
      ),
    ),

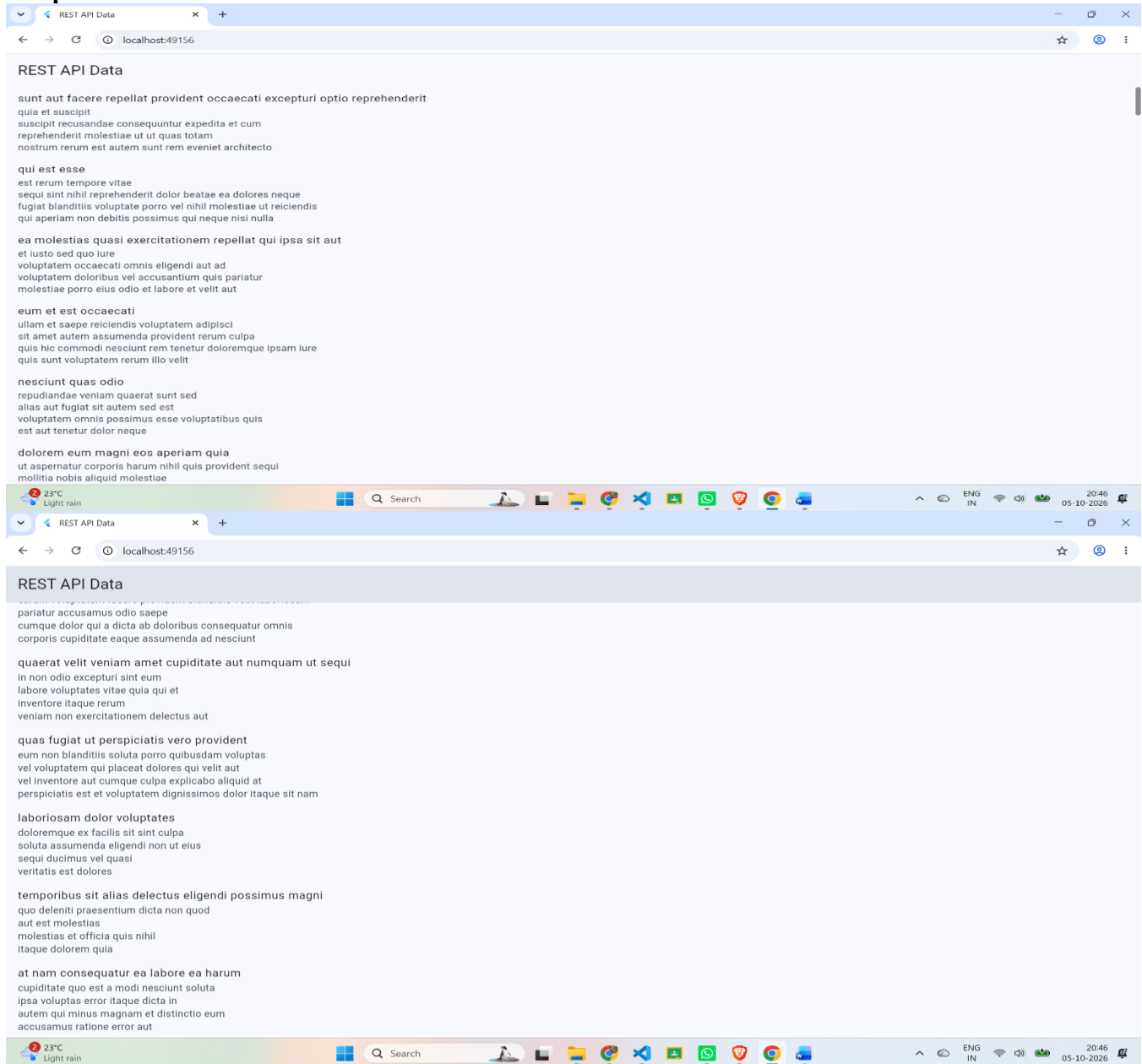
    body: isLoading
      ? const Center(
        child: CircularProgressIndicator(
          color: Color(0xFF9C7AC7),
        ),
      )
      : errorMessage.isNotEmpty
        ? Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              const Icon(
                Icons.error_outline,
                size: 60,
                color: Color(0xFF9C7AC7),
              ),
              const SizedBox(height: 15),

              Text(
                errorMessage,
                style: const TextStyle(
                  fontSize: 18,
                  color: Color(0xFF5E4A6E),
                ),
              ),

              const SizedBox(height: 15),
```

```
ElevatedButton(  
  onPressed: fetchData,  
  style: ElevatedButton.styleFrom(  
    backgroundColor:  
      const Color(0xFFBFA2DB),  
    foregroundColor: Colors.white,  
  ),  
  child: const Text('Retry'),  
),  
],  
,  
)  
: RefreshIndicator(  
  color: const Color(0xFF9C7AC7),  
  onRefresh: fetchData,  
  
  child: ListView.builder(  
    padding: const EdgeInsets.symmetric(  
      vertical: 10,  
    ),  
    itemCount: posts.length,  
  
    itemBuilder: (context, index) {  
      return Card(  
        child: ListTile(  
          contentPadding:  
            const EdgeInsets.all(12),  
  
          leading: CircleAvatar(  
            backgroundColor:  
              const Color(0xFFD8C4E8),  
  
            child: Text(  
              '${posts[index]['id']}',  
              style: const TextStyle(  
                color: Color(0xFF5E4A6E),  
                fontWeight: FontWeight.bold,  
              ),  
            ),  
          ),  
  
          title: Text(  
            posts[index]['title'],
```

```
style: const TextStyle(  
  color: Color(0xFF4D3A5C),  
  fontWeight: FontWeight.bold,  
  fontSize: 16,  
),  
),  
  
subtitle: Padding(  
  padding: const EdgeInsets.only(  
    top: 8,  
  ),  
  
  child: Text(  
    posts[index]['body'],  
    style: const TextStyle(  
      color: Color(0xFF6E5A7D),  
      height: 1.4,  
    ),  
  ),  
),  
),  
),  
);  
},  
),  
),  
);  
}  
}
```

Output:**RESULT**

Thus, The Flutter application successfully accesses a REST API using the HTTP package, retrieves JSON data from JSONPlaceholder, converts the response into Dart data, and displays the posts in a scrollable UI with loading and error handling.